

S Bharat Kumar Varma

Hyderabad, Telangana | bharatvarma6222@gmail.com | +91 9010079998

LinkedIn: linkedin.com/in/bharat-varma-6222tg | GitHub: github.com/bharatvarma6222-TG

PROFILE

AI/ML enthusiast with hands-on experience building LLM-powered applications, Retrieval-Augmented Generation (RAG) pipelines, and AI agents. Strong foundation in Python, embeddings, vector databases, and agent workflows. Seeking an AI/ML Internship in Agentic AI to contribute to real-world intelligent systems.

SKILLS

Languages & Frameworks: Python · FastAPI · LangChain · LangGraph

AI / ML: LLMs · RAG · Embeddings · Semantic Search · Prompt Engineering · Hugging Face

APIs & Platforms: OpenAI · Anthropic · LangSmith

Vector Databases: FAISS · Chroma

Databases: PostgreSQL · SQLite

DevOps & Tools: Docker · Linux · Git & GitHub · Jupyter Notebook

PROJECTS

BehaviorOS V4 – Autonomous Behavioral Intelligence & Cognitive HUD

- Built a real-time Windows behavioral analysis system using Win32 ctypes hooks (GetLastInputInfo, OpenInputDesktop) for OS-level activity telemetry, detecting idle, screen lock, and sleep states with a 3-second noise filter.
- Deployed a local semantic classification engine (sentence-transformers/all-MiniLM-L6-v2) to embed window titles and classify sessions into 10 behavioral categories (coding, debugging, research, deep work, entertainment, etc.) entirely offline.
- Engineered a LangGraph-driven daily briefing engine computing 13 behavioral metrics including longest focus streak, context-switching rate, and distraction frequency from raw SQLite telemetry logs.
- Built a PyQt6 glassmorphic HUD overlay with a LeetCode-style memory heatmap calendar, live focus dials, and cognitive state sparklines, backed by an async pub/sub event bus and FastAPI gateway.

Local SLM App with Ollama – Offline Model Benchmarking

- Deployed three small language models entirely offline using Ollama, eliminating cloud API dependency and addressing privacy and data-residency constraints.
- Benchmarked inference performance measuring tokens/sec, first-token latency, and memory footprint across all three models on identical hardware.
- Documented quality-vs-speed tradeoffs to guide model selection based on latency budgets and task complexity.

Offline ChatGPT Clone – Fully Local AI Assistant

- Built a fully offline AI chat app using FastAPI, Streamlit, and Ollama (Qwen2.5:3b) with no cloud dependency.
- Implemented streaming responses and persistent multi-conversation history via SQLite and SQLAlchemy.
- Integrated LangChain RAG pipeline supporting PDF upload and local vector retrieval for context-aware responses.

EDUCATION

Bachelor of Technology (B.Tech) – Computer Science

Parul University | Expected Graduation: 2027

LEARNING & EXPERIENCE

- Hands-on experience building LLM-powered applications using Hugging Face and LangChain.
- Implemented RAG pipelines with FAISS and Chroma embedding-based retrieval.
- Built AI agents with tool-calling and reasoning workflows through project-based learning.
- Strong understanding of semantic search, cost optimization, and real-time data access.
- Familiar with containerization using Docker and Unix/Linux environments for development and deployment.